

OpsBrain — Unified Industrial Knowledge Intelligence

Official Submission Document · ET AI Hackathon 2026 · Problem Statement #8: Industrial Intelligence

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Target Industry: Heavy Industry & Refineries	Repository: github.com/saurabhmj11/OPSBRAIN	Status: Production Ready (18 Docs Ingested)

1. Executive Summary & Problem Context

In asset-intensive industrial environments—such as petroleum refineries, chemical processing facilities, power plants, and manufacturing sites—unplanned downtime costs plant operators millions of dollars per hour. The root cause is rarely an absence of technical documentation; rather, it is **industrial knowledge fragmentation**.

Industrial engineering data is fragmented across **7 to 12 disconnected document systems** (NASSCOM-EY). Field engineers spend up to **35% of working hours** searching for information or recreating lost procedures (McKinsey). Furthermore, **18–22% of unplanned downtime** in heavy industry is driven by maintenance teams acting without full equipment history (BIS Research), while **~25% of experienced industrial engineers** retire within the decade, taking undocumented tacit knowledge with them.

***The Core Industrial Bet:** Industrial knowledge loss is not a search problem—it is a **connection** problem. The solution is not a generic chatbot over a PDF folder, but an entity-linked knowledge layer that enables reasoning across work orders, inspection logs, OEM manuals, and safety standards.*

2. Target User Personas & Value Proposition

Persona	Operational Context	Primary Value Delivered
Field Technician (Raju)	On-site, mobile device, low connectivity.	Instant answers to 'Has valve V-118 failed before?' with verified citations on m
Maintenance Engineer (Priya)	Desktop, plans plant maintenance schedules.	Cross-document root-cause analysis across work orders, OEM manuals & ins
Compliance Officer (Anil)	Prepares for OISD-118 / Factories Act audits.	Automated gap analysis diffing plant procedures against regulatory standards
Plant Manager (Meena)	Oversees plant safety & uptime KPIs.	Macro visibility into recurring equipment failure risks and compliance coverage

3. System Architecture & Technical Pipeline

OpsBrain employs a multi-tier hybrid architecture combining dense vector embeddings with structured entity-graph traversal:

Pipeline Layer	Technical Mechanism	Operational Output
1. Ingestion Layer	Multi-format parser for OEM manuals, work orders, P&IDs, SOPs, inspection logs, etc.	Ingested documents split into 1536-dim vector chunks.

2. Entity Graph Index	LLM-assisted NER and relation extraction (Equipment, Personnel, Work Orders, Clauses). Graph with 7 entity node types.	Cross-Work Order Clauses
3. Hybrid Retrieval Engine	Dense vector similarity search combined with 2-hop graph nodes. Ranker creates candidate set with graph entity metadata.	Ranker
4. Agent Layer	Citation-first synthesis, proactive incident surfacing, and statutory compliance. Verified answers with source chips, similarity alerts, and SOP drawer.	Verified answers

4. Flagship Autonomous Agents

A. Expert Knowledge Copilot (Citation-First RAG)

- **Mandatory Citations:** Every factual claim in an answer is tied to an explicit source marker [C1], [C2].
- **Fail-Closed Confidence Gate:** If confidence falls below 18%, the copilot explicitly states *'Insufficient evidence in ingested corpus'* rather than guessing.
- **Multi-Stage Reasoning Visualizer:** Shows real-time execution steps: *Vector Search @ 1536-dim → Graph Expansion → Citation Synthesis*.
- **1-Click Copy & Drawer:** Supports 1-click answer copying and direct source drawer preview.

B. Proactive Lessons Learned Agent (Incident Surfacing)

- **Incident Similarity Surfacing:** Compares active work order descriptions (e.g. *'WO-4710: Compressor C-302 tripped on high vibration'*) against historical incident logs.
- **Scenario Presets:** 1-click test triggers for major plant events (Seal leaks on Pump P-204, Surge on C-302, Valve V-118 blowout).
- **Actionable Recommendations:** Surfaces match percentages, agent rationale, and recommends escalation without auto-issuing destructive directives.

C. Compliance Command Center (Statutory Audit)

- **Statutory Standard Mapping:** Audits plant SOPs against OISD-118 and Factories Act 1948 regulations.
- **Plant Compliance Score:** Visual gauge showing Covered vs. Partial vs. Uncovered Gap clauses.
- **Side-by-Side Clause Diff:** Displays mandatory regulatory clause text next to extracted plant procedure evidence, with a 1-click *'Draft SOP Revision'* trigger.

5. Knowledge Graph Model & Data Schemas

OpsBrain ingests 18 heterogeneous plant documents, extracting 7 distinct entity types and linking them via typed graph relations:

Entity Type	Example Identifiers	Graph Relations
Equipment	P-204 (Crude Pump), C-302 (Compressor), V-118 (Valve)	FAILED_ON, MAINTAINED_BY, GOVERNED_BY
Work Order	WO-4521, WO-4710, WO-4712	TARGETS, PERFORMED_BY, REFERENCED_IN
Incident / Near-Miss	INC-2024-031, NM-2024-045	INVOLVES, RESOLVED_BY, APPLIES_TO
Procedure / SOP	SOP-PUMP-004, PR-MECH-012	COVERS_CLAUSE, MENTIONS_EQUIPMENT
Regulatory Clause	OISD-118:4.3.2, FactoryAct-1948:Sec31	GOVERNS_PROCEDURE, MANDATES_AUDIT
Personnel	Tech S. Kumar, Inspector M. Patel	PERFORMED, AUTHORED, ACKNOWLEDGED

6. Agent Behavior Guardrails & Security Rules

- **Non-Negotiable Citations:** Answers without source claims are automatically stripped before reaching user UI.
- **Fail Closed Safety:** Low confidence answers (>0.18 threshold) return explicit non-answers rather than hallucinated guesses.
- **Paraphrase vs Direct Quote Budget:** Verbatim quotes are reserved only for safety-critical instructions or regulatory clauses.
- **Scope Discipline:** Agents operate within fixed tool surfaces and communicate via structured JSON evidence objects.

7. Tech Stack & System Verification

- **Web Framework:** Next.js 16 (App Router), React 19, TypeScript 5
- **UI & Styling:** Tailwind CSS v4, Lucide Icons, Sonner Notifications, Glassmorphism utilities
- **Data & ORM Layer:** Prisma ORM (SQLite / PostgreSQL) initialized with dev.db
- **Verification Status:** 100% verified, listening live on http://localhost:3000

8. Submission Sign-off & Team Credits

OpsBrain successfully demonstrates how combining dense vector search with structured knowledge graphs transforms fragmented technical documentation into verifiable, actionable industrial intelligence.

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